

Cary Jones

Python Software Engineer | Test Automation, Data Systems & Device Integration

Temple City, CA 91780 (Los Angeles area) (626) 272-2567 cgjonesdev@gmail.com
linkedin.com/in/cary-g-jones github.com/cgjonesdev cgjonesdev.github.io/carygjones

SUMMARY

Python Software Engineer with 15+ years building production libraries, REST APIs, and automated test infrastructure. Experienced designing **data aggregation pipelines**, extensible internal frameworks, and **Docker**-containerized services on AWS. Five years at Hospira verifying embedded medical device software — requirements-based testing, hospital network integration, and hardware-adjacent debug workflows in regulated environments. Strong networking fundamentals (U.S. Marine Corps communications specialist) and hands-on work with PostgreSQL, Redis, MongoDB, socket-driven integrations, and CI/CD test automation.

CORE COMPETENCIES

Python: Library and framework design, pytest, FastAPI, Django, Flask, Asyncio, Celery, Pydantic, packaging reusable modules for engineering teams

Test & Automation: Automated test suites, equipment/driver-style integrations, debug tooling, socket and API interfaces, CI/CD gates, regression triage workflows

Data Systems: PostgreSQL, MySQL, MongoDB, Redis — schema design, aggregation pipelines, query optimization, event-driven ETL at scale

Infrastructure: Docker, AWS ECS/Fargate, Lambda, CloudWatch, Linux, GitHub Actions, Secrets Manager, networking (TCP/IP, TLS, VPN concepts)

KEY IMPACT

- Built reusable **Python test and deployment frameworks** with automated pytest suites and pre-deploy CI/CD validation — reducing production defects and merge conflicts by **70%**.
- Engineered event-driven data pipelines on **AWS ECS/Fargate** aggregating **2.5M+ daily transactions** at **99.95% accuracy** across PostgreSQL, Redis, and external APIs.
- Architected secure Python REST APIs and relational data models for healthcare platforms serving **500k+ user profiles** with OAuth2 and encryption hardening.
- Deployed containerized Python microservices achieving API latency under **120ms** and **+45% throughput** improvement.
- Five years verifying **embedded device software** and hospital network integrations for regulated infusion pump systems — formal verification, protocol testing, and cross-team triage.

PROFESSIONAL EXPERIENCE

Principal Software Engineering Consultant / Technical Lead

Jan 2023 – Present | Los Angeles, CA / Remote

CGJSoftware, Inc.

- Developed extensible **Python libraries and microservice frameworks** (FastAPI, Django) used across client projects — standardized API patterns, pytest harnesses, and Docker deployment templates.
 - Designed data aggregation and caching layers (PostgreSQL, Redis) for high-read API endpoints; built monitoring and debug tooling with CloudWatch and structured logging.
 - Implemented CI/CD pipelines with automated integration tests, container builds, and staged rollouts on **AWS ECS/Fargate**.
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API Developer, System Architect, DevOps Engineer, and Scrum Master

January 2018 – July 2019 | Alan Health & Science Corp. (Remote)

- Built Python REST APIs and internal tooling for HIPAA-compliant platforms serving **500k+ active user profiles**.
 - Designed PostgreSQL schemas and data aggregation workflows; established Git branching standards and automated pre-deploy test scripts.
 - Led cross-functional team as Scrum Master; improved release predictability and test coverage across the stack.
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Senior Software Engineer

October 2021 – November 2022 | Disney Corporation, Inc. (Remote)

- Engineered Python backend modules and **PostgreSQL data interfaces** for financial systems processing **\$10M+ daily metrics** at **99.99% uptime**.
 - Co-architected automated costing and reconciliation tooling — cut batch runtime from 8 hours to 45 minutes through optimized aggregation logic.
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Quality Assurance & Network Integration Engineer

February 2005 – December 2009 | Hospira, Inc., San Diego, CA

- Five years in **regulated device verification** — automated and manual test frameworks, hospital network integration, firmware/hardware coordination, and formal requirements traceability.
 - Debugged protocol-level issues across embedded systems and external interfaces; documented reproducible test procedures for engineering triage.
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Python API Developer

August 2013 – April 2014 | Fidelity Investments, Irving, TX

- Built **Django REST Framework** APIs for datacenter network automation — reduced provisioning from **4 hours to 10 minutes** with robust contract design and pytest coverage.
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EDUCATION

Associate of Science, Computer Electronics

ITT Technical Institute — GPA 3.67

United States Marine Corps

Field Wireman & Communications Specialist — Honorably Discharged